

Dynamic Mesh Sharpening

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Figure 1: Dynamically sharpened Wolf’s cape from the DreamWorks film *Puss in Boots: The Last Wish*

ABSTRACT

Dynamic mesh sharpening is a custom new process used throughout the film *Puss in Boots: The Last Wish* to dynamically flatten out cloth surfaces and sharpen folds/wrinkles to impart a more graphic planar look than the look obtained from the cloth simulation alone. This process allowed the planar look to be realized in animated shots rather than just in a reference sculpted model. It allowed us to see simulated clothing with sharpened wrinkles that had the same design language as the rest of the film assets. This talk presents the comprehensive approach used to dynamically sharpen meshes.

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1 MOTIVATION

Previous films at DreamWorks Animation have called for soft and rounded wrinkles on the simulated clothing with most of the variation being based on design, cut, and movement happening around the properties of the clothing rather than the design of the wrinkles and folds. The design language for the film *Puss in Boots: The Last Wish* was based on a simple, planar, and graphic look. All soft edges were made sharp, including everything from props to fingers and faces. The simulated cloth wrinkles needed to live in the same design language [Figure 1] as the sharper and planar static models.

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2 PRIOR WORK AND INITIAL ATTEMPTS

Existing tools in third-party applications try to achieve a similar look using only a geometric sharpening approach by moving 3-D point positions, but are limited to sculpting/modeling, and cannot be used on animated geometry. In those tools 1) the geometric sharpening approach will lead to artifacts discussed in Section 3.2, 2) creating extremely sharp meshes can be difficult and unstable, 3) sharpness convergence is slow, and 4) temporal coherence, volume preservation, and collision avoidance are not taken into account.

One of our earlier attempts involved modeling the sharp folds on the render mesh at reference pose. This pre-modeled sharp wrinkle was present all the time during simulation and was not dynamic. It looked like cement draped on flowing/baggy garments rather than clothing ebbing and flowing with folds and wrinkles. Another approach was to simulate a low resolution mesh with some post processing, but even though it looked planar, the look resembled decades-old video games. An attempt was made to simulate a high resolution mesh and control the folds and wrinkles with simulation settings. The simulation time was costly in this method and did not produce the desired sharpened look. We tried reducing the high resolution of the mesh after simulation to get a planar look, but that was not stable and did not produce the desired graphic look.

3 DYNAMIC MESH SHARPENING

The approach that we finally used was a comprehensive one taking into account the geometric, motion, and rendering aspects of creating a dynamic sharpened mesh. The major two steps in our approach are *Geometric Sharpening* and *Normal Sharpening*.

3.1 Geometric Sharpening

Geometric Sharpening involves the process of moving vertices of the mesh in 3-D to produce the sharpened look. It is a multi-step process that allows the artist control, using parameters/attributes, at every step. Those steps are described next.

- The sharpening needs to emphasize the larger folds in the cloth in order to provide the desired graphic look. The mesh is pre-processed to remove smaller parts like buttons, and to smooth out tiny folds and spatial noise in the simulation. An option to offset the mesh along its normals is provided. The mesh is subdivided to provide enough vertices to be able to sharpen geometrically.
- The mean curvature of the mesh is calculated at its vertices.
- Mesh vertices are iteratively displaced such that vertices near high curvature regions move towards the higher curvature regions to increase sharpness, and vertices in the lower curvature regions are locally smoothed out to contrast the sharpening even further. The artist can also locally control the sharpness by painting point scaling attributes. In each iteration, the effect of sharpening and smoothing can be modulated. So, for stability, the artist has control to run several iterations of this process with incremental sharpening to attain the final sharpened result.
- A projection step is provided as part of the iteration to help with retaining volume. One mode of projection involves projecting the results to the closest point on the original mesh $Mesh_{orig}$ at each step of the iteration. In the second mode of projection, for each vertex V from the mesh $Mesh_{geom_sharp}$, with position Pos_{geom_sharp} obtained after a sharpening iteration, the closest point P_{cl} on $Mesh_{orig}$ is found, P_{cl} 's polygonal face F_{cl} and parametric UV (UV_{cl}) on that face are calculated, and F_{cl} and UV_{cl} are used to calculate the new position $Pos_{geom_sharp_proj}$ of vertex V on $Mesh_{geom_sharp}$. This provides faster convergence of the sharpening and the ability to create extremely sharp meshes. This projection is followed by an optional smoothing step as part of the iteration to remove crunching artifacts if any.
- A mesh relaxation option (to relax vertices along the surface) is provided to reduce the wide variations in the layout of the topology, if any, after sharpening. The incremental and iterative sharpening, projection, and smoothing steps, along with mesh relaxation provide stability and temporal coherence. A collision resolution module is provided to resolve collisions with an object.

3.2 Normal Sharpening

The application of texture to the geometrically sharpened mesh exposed the artifact of texture swimming while the sharpened points moved around to keep a sharpened look of the dynamic folds and wrinkles in a simulated cloth. A simple closest point projection of texture UV coordinates from the original mesh to the sharpened mesh did not work because of the discontinuities at the texture UV border. So, an approach that transferred normals from the geometric sharpened mesh to the original mesh using normal maps was used. The steps in that process are described next.

- Obtain the geometric sharpened mesh $Mesh_{geom_sharp}$.
- Create a flat point cloud $P_{for_N_map}$ on a grid (1K or 2K res).
- Using texture UV, project $P_{for_N_map}$ onto $Mesh_{orig}$ to take that mesh's 3-D shape. Obtain the normals of $P_{for_N_map}$ from $Mesh_{geom_sharp}$ by projecting them onto $Mesh_{geom_sharp}$. [Figure 2]. Convert $P_{for_N_map}$ to a normal map, Map_{sharp_normal} .
- Combine Map_{sharp_normal} with any existing normal maps using the *CombineNormalMap* shader (which uses *Reoriented Normal Mapping* [Barré-Brisebois and Hill 2012]), and apply the combined normal map to $Mesh_{orig}$ in the shader at render time.

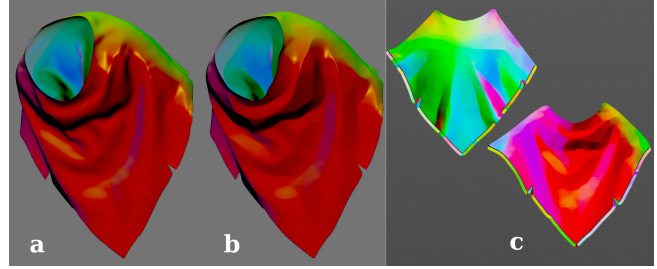


Figure 2: a) Original mesh, b) Sharpened mesh, and c) Point cloud in texture space with normals obtained from sharpened mesh in 3-D. All are visualized with normals as colors.

As only the original simulated mesh is finally rendered, the *Normal Sharpening* step removes the texture artifacts, as well as any collision issues, and the subdivided sharpened mesh need not be exported out for render.

4 IMPLEMENTATION

The Sharpening process is implemented using custom nodes in a third-party package. The *Geometric Sharpen* node performs the *Geometric Sharpening* step which takes the original mesh as input. This node provides the artist with various parameters for sharpening, including pre and post operations, along with allowing the artist to locally control sharpening geometrically. The *Normal Sharpen* node performs the *Normal Sharpening* step which takes the original mesh and sharpened mesh as inputs. Normal map options such as map resolution and projection types, and preview options such as previewing the point cloud in 3-D space (with projected normals) or in 2-D texture space [Figure 2] are provided by this node. The normal map output from this node is written out to disk.

The sharpening process, which is run after the cloth simulation, is repeated per UDIM, per material, per subasset, and for every frame as the folds and wrinkles change in the cloth simulation, each time producing a new normal map. The process is procedurally batched and run in parallel. The normal maps are procedurally injected into character material networks using USD. A shared USD file is used to set up the sharpened normal shaders.

5 RESULTS AND CONCLUSION

Dynamic mesh sharpening was used on the film *Puss in Boots: The Last Wish* on hero and generic garments. This comprehensive mesh sharpening approach was tested on several simulated garments and modified to fit artistic direction, and maintain control and stability. All *Character Effects* artists on the film used mesh sharpening, and it was used on $\approx 75\%$ of the shots of the film. It allowed for sharpened styles on dynamic assets and hence allowed the simulated clothing to be in the same design language as the rest of the assets. It provided the ability to retain stable texturing during dynamic stylization. It paved the way for stylizations using geometric post-processing and material/shader deliveries from *Character Effects*.

REFERENCES

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